Editor-in-Chief
Discover Mental Health

From:
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On behalf of the Corresponding Author:
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Dear Editor-in-Chief,

On behalf of my co-author and corresponding author, Professor Md. Jamal Uddin, PhD, we are pleased to resubmit our revised manuscript entitled:

"**Digital Connectivity and Mental Health Symptoms Among Women in Bangladesh: A Rural-Urban Perspective**"

for reconsideration in ***Discover Mental Health*** following the minor revision decision and valuable comments provided by the Editor and reviewers.

We sincerely appreciate the Editor and reviewers for their constructive feedback, which has helped us improve the clarity, methodological transparency, and overall quality of the manuscript. We have carefully addressed all comments and revised the manuscript accordingly.

In response to the reviewers' suggestions, we have undertaken the following major revisions:

- Updated tables and figures to improve clarity, readability, and presentation quality.
- Added relevant references and strengthened the background and methodological justification.
- Performed comprehensive language editing and improved the overall scientific presentation of the manuscript.
- Revised the interpretation of findings, terminology, and limitations to ensure appropriate and cautious conclusions.
- Improved the reproducibility of the study by providing all analysis codes and relevant materials in our GitHub repository:

  https://github.com/muhammadsalek/digital-connectivity-mental-health-bangladesh-women/blob/main/README.md

For transparency and ease of evaluation, we have submitted the following files:

1. Clean revised manuscript (final version)
2. Revised manuscript with tracked changes/highlighted modifications (all newly added or modified text is highlighted in green)
3. Point-by-point response to reviewers

The revised manuscript includes page numbers and line numbers corresponding to the changes made. We believe that these revisions have substantially strengthened the scientific rigor, clarity, and reproducibility of our study.

We sincerely appreciate your time and consideration of our revised manuscript and look forward to your further evaluation.

Sincerely,

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Major Comments

Reviewer Comment 1: Conceptualization of digital connectivity

The manuscript combines internet use, ICT ownership, and ICT usage under the broad concept of "digital connectivity." However, these variables represent different constructs. Internet use reflects individual engagement with digital platforms, whereas ICT ownership mainly reflects household access to communication devices. Similarly, the ICT usage variable is largely based on exposure to traditional media such as television, radio, and newspapers. The rationale for combining these indicators within a single conceptual framework needs to be strengthened. A clearer explanation of how each exposure relates to mental health would improve the overall coherence of the study.

Response to Reviewer [Pages no:3-4, line no:64-74, page no:8, line no:161-180, page no: 25, line no: 540-542]

We sincerely thank the reviewer for this valuable comment. In the revised manuscript, we have clarified the conceptual framework and rationale for including these indicators.

Specifically, internet use represents individual-level engagement with digital platforms, ICT ownership reflects household-level access to communication technologies (e.g., television, radio, mobile phone, and computer), and ICT usage represents exposure to information sources such as television, radio, and newspapers. These indicators were analyzed separately to capture different but complementary dimensions of digital connectivity rather than combining them into a single index.

The exposure measures were constructed using standardized BDHS 2022 questionnaire items and informed by previous DHS-based studies. Previous research has used household ownership of communication technologies as a proxy measure of ICT access in population-based surveys (Cardoso & Sorenson, 2017; Nandi et al., 2024). Relevant references and explanations have now been added to the Methods section.

We also acknowledge that ICT ownership does not capture individual-level patterns of technology use, including frequency, purpose, and intensity of engagement. Additionally, equal weighting of different devices may not fully represent their functional differences. These measurement limitations have now been explicitly discussed in the revised manuscript.

Changes made:

- Clarified the conceptual distinction between internet use, ICT ownership, and ICT usage.
- Added supporting references from previous DHS-based studies (Cardoso & Sorenson, 2017; Nandi et al., 2024).
- Clarified that ICT ownership represents household-level digital access rather than device-specific effects.
- Expanded the limitations section regarding measurement constraints, including the lack of detailed information on frequency, quality, and purpose of digital engagement.

Reviewer Comment 2: Missing data and internet-use analysis

The manuscript indicates that only 5,486 women were included in the internet-use models compared with nearly 19,000 women in the main dataset. This substantial reduction in sample size raises concerns about potential selection bias. The manuscript should clarify why internet-use data were unavailable and whether women included in the analysis differed systematically from those excluded.

Response: [Page no: 7, line no: 144-145, Page no: 24-25, line no:532-535, page no: 20, line no:437-438]

We sincerely thank the reviewer for highlighting this important issue. We would like to clarify that the reduced sample size was not due to missing internet-use data alone; rather, the internet-use models were conducted using complete-case analysis among eligible respondents with available information on internet use and all required covariates. Participants without complete information for the specific model were excluded from that analysis, resulting in a smaller analytical sample (n = 5,486).

We have now clarified this analytical approach in the Methods section and acknowledged that the reduced sample size may introduce selection bias and limit the generalizability and statistical power of the internet-related findings. This limitation has been added to the revised manuscript.

Changes made:

- Clarified that the internet-use analysis was based on complete-case analysis among eligible participants.
- Explained the reason for the reduced analytical sample size.

- Added a limitation regarding potential selection bias and reduced generalizability of internet-related findings.

Reviewer Comment 3: Confounder selection and model development

The description of confounder selection is difficult to follow. The manuscript states that covariates were selected using a DAG, theoretical considerations, literature review, and statistical significance testing. It is not clear how these different approaches were combined in practice.

A clearer and more transparent explanation of the model-building process is needed. At present, readers may find it difficult to understand why certain variables were included while others were excluded.

Response: [Page no: 10, line no:213-223]

We sincerely thank the reviewer for this important comment. We agree that the previous description of confounder selection was not sufficiently clear. We have revised the Methods section to clarify the model-building approach.

In this study, potential confounders were primarily selected a priori based on the directed acyclic graph (DAG), previous literature, and theoretical understanding of factors related to digital connectivity and mental health symptoms. The DAG approach was used to identify the minimally sufficient adjustment set to reduce potential confounding bias.

The chi-square test was conducted only as an exploratory assessment of crude associations between variables and was not used as the sole criterion for confounder selection. Statistical significance was considered alongside theoretical relevance and the conceptual framework when developing the final multivariable models.

We have revised the manuscript to clarify that confounder selection was primarily theory-driven rather than based solely on statistical significance.

Changes made:

- Clarified that DAG, literature review, and theoretical considerations were the primary basis for confounder selection.
- Clarified that chi-square testing was used only for descriptive/exploratory assessment.
- Removed ambiguity regarding the role of p-values in covariate selection.
- Revised the Methods section to provide a clearer model development process.

Reviewer Comment 4: Interpretation of rural–urban differences

One of the main conclusions of the manuscript is that there are important rural–urban differences in the relationship between digital exposure and mental health. However, the analyses rely on

separate rural and urban models without formally testing interaction effects. Without a statistical test of interaction, it is difficult to conclude that associations truly differ between rural and urban women.

Response: [Page no: 16, line no:344-349, page no:25, line no:544-547]

We sincerely thank the reviewer for this important observation. We agree that separate rural and urban stratified analyses do not formally establish effect modification. Therefore, we have revised the interpretation of our findings to avoid implying statistical differences between rural and urban groups.

We have clarified in the revised manuscript that the rural–urban stratified findings should be interpreted as descriptive differences in associations rather than confirmed effect modification. We have also added this issue to the limitations section and acknowledged that future studies incorporating formal interaction analyses are needed to confirm whether rural–urban residence modifies the association between digital connectivity and mental health symptoms.

Changes made:

- Revised the interpretation of rural–urban differences to avoid overstatement.
- Added a limitation regarding the absence of formal interaction testing.
- Clarified that stratified analyses represent descriptive comparisons rather than confirmed effect modification.

Reviewer Comment 5: Spatial analysis

The spatial component of the study is visually appealing but remains largely descriptive. The maps illustrate regional variation, yet no formal spatial analysis appears to have been conducted. Consequently, statements regarding spatial inequalities and geographic clustering should be interpreted carefully.

Response: [Page no:17, line no:361-366, page no: 18, line no:379-384, page no:25, line no: 548-549, page no:26, line no:558-559]

We sincerely thank the reviewer for this valuable comment. We agree that the spatial component of our study is primarily descriptive and does not provide evidence of formal geographic clustering or spatial effects.

Accordingly, we have revised the interpretation of the spatial findings throughout the manuscript to describe regional variation rather than implying spatial clustering or geographic effects. We have also clarified in the limitations section that formal spatial methods, such as spatial autocorrelation analysis (e.g., Moran's I) and spatial regression approaches, were not performed in this study.

We acknowledge that future research incorporating advanced spatial analytical methods would provide further insight into geographic patterns and potential spatial determinants of digital connectivity and mental health symptoms.

Changes made:

- Revised spatial interpretations to emphasize descriptive regional variation.
- Removed statements implying geographic clustering or spatial effects.
- Added limitations regarding the absence of formal spatial analyses.
- Suggested future studies using spatial autocorrelation and spatial regression approaches.

Minor Comment

Reviewer Minor Comment 1: Language editing

Comment:
The manuscript would benefit from substantial language editing. Numerous sentences are difficult to follow, particularly in the Introduction and Discussion sections.

Response: [Thought Manuscripts]

We sincerely thank the reviewer for this helpful suggestion. We have carefully revised the manuscript and performed comprehensive language editing throughout the Introduction, Discussion, and other sections to improve clarity, readability, and scientific presentation.

Changes made:

- Revised unclear and complex sentences throughout the manuscript.
- Improved grammar, sentence structure, and overall readability.

Reviewer Minor Comment 2: Terminology of depression and anxiety

Comment:
The terms "depression" and "anxiety" are used throughout the paper, although PHQ-9 and GAD-7 identify symptoms rather than clinical diagnoses. More precise terminology should be used.

Response: [Thought Manuscript]

We sincerely appreciate the reviewer's important observation. We agree that PHQ-9 and GAD-7 assess depressive and anxiety symptoms rather than clinical diagnoses. Accordingly, we have revised the terminology throughout the manuscript and replaced inappropriate use of "depression" and "anxiety" with "depressive symptoms" and "anxiety symptoms" where appropriate.

Changes made:

- Updated terminology throughout the manuscript to accurately reflect symptom-based assessment using PHQ-9 and GAD-7.

Reviewer Minor Comment 3: Repetition in Introduction and Discussion

Comment:
There is considerable repetition in both the Introduction and Discussion.

Response:
We sincerely thank the reviewer for this valuable suggestion. We have carefully reviewed the Introduction and Discussion sections and removed repetitive statements while improving the flow and logical organization of the manuscript.

Changes made:

- Reduced unnecessary repetition.
- Improved the structure and coherence of the Introduction and Discussion sections.

Reviewer Minor Comment 4: Tables and Figures readability

Comment:
Several tables and figures could be simplified to improve readability.

Response: [page no: 35-42, line no:770-796]

We appreciate the reviewer's helpful suggestion. We have revised the tables and figures to improve readability and presentation. Several elements were simplified, unnecessary details were removed, and figure/table formats were updated for better interpretation.

Changes made:

- Simplified tables and figures.
- Improved formatting and visual clarity for better reader understanding.

Closing Statement

We sincerely appreciate the Editor and reviewers for their valuable time and constructive feedback. We believe that the revisions made have improved the clarity, rigor, and reproducibility of our manuscript. We kindly request your consideration of the revised manuscript for publication in ***Discover Mental Health*.**